Data Architect Interventions

EMPOWERING High Performance Technology Teams

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About StackRoute ©



Founded in 2015 as a product engineering startup, StackRoute provides disruptive learning solutions to produce world class technology professionals with deep skills and competence.

StackRoute aspires to be a preferred learning partner for organizations enabling and adopting digital transformation

StackRoute operates with a singular belief - to build business-aligned talent capabilities. This means:

- Driving learning outcomes that create demonstrable skills
- Delivering learning effectiveness at scale

GSIs

GCCs

Consulting

Analytics and AI

I never teach my pupils.
I only attempt to provide the condition
In which they can learn
- Albert Einstein

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StackRoute: Solutions Portfolio ©

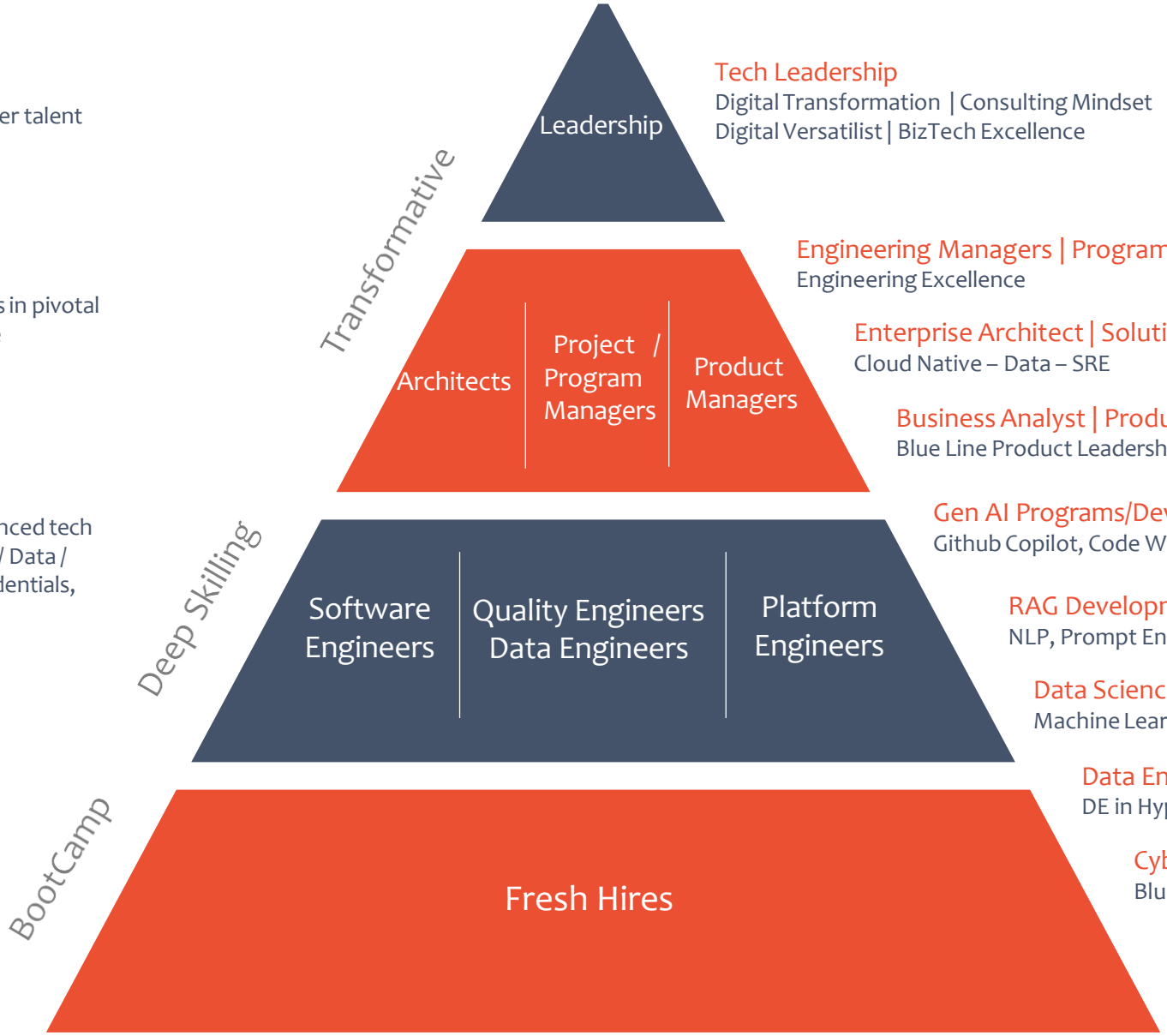
Learning Design and Talent Advisory
Designing learning and assessment to deliver talent capability aligned with business needs

Transformative Programs
Transforming experienced tech professionals in pivotal roles to lead digital projects and deliver value

Deep Skilling Programs
Re Skilling / Up skilling / Cross skilling experienced tech professionals into modern Full Stack / Digital / Data / Cloud / Platform / AI Professionals; Micro credentials, Flexible Learning Models

Unique Multi-stage Assessment
Calibrates candidate's Knowledge and Skill quotient

Immersive/Virtual Programs
Transforms fresh hires into full stack Product Engineers / Software Engineers / Data Engineers / Site Reliability Engineers



Tech Leadership
Digital Transformation | Consulting Mindset
Digital Versatilist | BizTech Excellence

Engineering Managers | Program Managers | Project Managers
Engineering Excellence

Enterprise Architect | Solution Architect | Application Architect
Cloud Native – Data – SRE

Business Analyst | Product Owner | Digital Product Managers
Blue Line Product Leadership

Gen AI Programs/Developer Tools | LLMs/SLMs | Cloud AI Stacks
Github Copilot, Code Whisperer, Duet, Cursor, Windsurf, Lllama, DeepSeek

RAG Development, Agentic AI
NLP, Prompt Engg, LangChain, Vector db

Data Science, AI/ML and Data Analytics
Machine Learning | Deep Learning | Data Visualization

Data Engineering
DE in Hyperscalers | Databricks | Snowflakes

Cybersecurity
Blue Team | Red Team | Purple Team

Platform Engineering
SRE | DevOps | Developers | Cloud Engg

Low Code No Code Development
Power Apps | Pega | Appian

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StackRoute Transformative Programs ©

Architect Competence Development	Product Management | Product Leadership	Program |Project | Delivery Excellence	Business Analyst | Product Owner	Tech Leadership	Digital Versatilist | Leading Digital Transformation
• Aspiring Architect • Architecting for Performance • Architecting for Change • Enterprise Architect	• Product Mgmt. Essentials • Digital Product Manager • Product Leadership	• Project Mgmt. Excellence • The Path to Program Mgmt. Excellence • Delivery Manager • EMPower: Engg. Manager • Toolkit for Tech Leadership	• Business Analyst • Automation Business Analyst • Product Owner	• Building a consulting Acumen • Smart Decision Making • Agile Excellence Practitioner • Orthogonal Paradigms • Data led decision making • Learning Compass	• Purpose driven mindful "Digital Versatilist" • Leading DT for Senior Leaders • Enabling DT for Mid to Senior-level Professionals • Enhancing Tech Competencies for Young Professionals

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Our Approach ©

Capability-led organizations require "Transformative Competencies"



Create Value

- Challenge the status-quo
- Collaborate with others
- Think "outside the box"



Reconcile Dilemmas

- Understand opposing positions
- Develop arguments
- Find practical solutions to conflicts

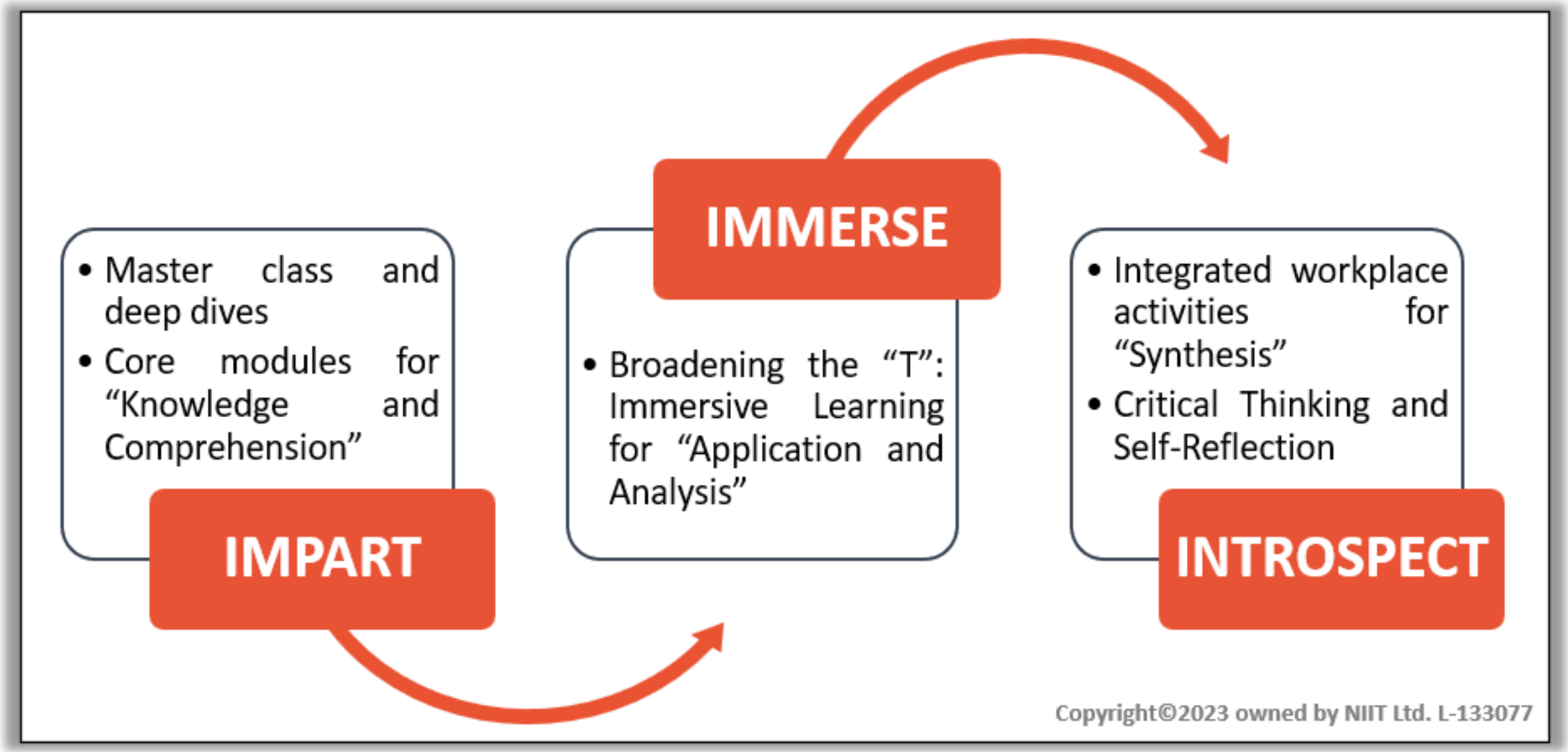


Taking Responsibility

- Reflect upon self
- Evaluate own actions
- Consider ethical and societal goals

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StackRoute's 3i Learning Framework©



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Awards and Recognitions

Winner of 15 Brandon Hall Group Excellence Awards
Jointly with our client partners
7 Gold | 3 Silver | 5 Bronze

STACK ROUTE AN NIIT VENTURE

Transformative Programs

Architecture Competence



Dell - TechEdge 24 for Architects



DXC - Aspiring Architect Program



Happiest Minds - Architecting for Change



Persistent - Architect Competence Development Program



TEKsystems - Aspiring Architect Upskilling Program



Tredence - Aspiring Data architect- Level 1

Engineering Excellence



Happiest Minds - LEAD Program - for Delivery Managers

Business Analysts and Product Owners (BAPO)



Happiest Minds - S.H.I.F.T. Program for Business Analysts

Deep-skilling Programs

Full Stack



CGI - Graduate Transformation Program (Future Talent Training Program)



Virtusa - Full Stack Development Program (Java and .NET)



Natwest - Talent Upskilling Strategy



Lloyds - Multiple Programs in Deep Skilling

Data Science



Tredence - Databricks Upskilling Initiative (DE Professional & GenAI)



Tiger Analytics - Data Engineering Bootcamp



Ness- Certified Data Engineer' Program

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Design 1A:

Aspiring Data Architect Level 1 for Senior Data Engineers

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Context: Level 1 – Aspiring Data Architect

Eligibility

- Good Understanding of the Data types (Master, Ref, Txn & Analytical Data)
- Types of DBMS systems (OLTP, OLAP/DSS, ODS)
- Data access patterns (Singleton Inserts, ACID, Analytical queries)
- Data Modelling fundamentals (ER / Dimensional)
- Normalization, Denormalization, Normal forms (Codd's Rules)
- Good Knowledge on DWH/Data Marts and ODS
- ETL patterns (ETL vs ELT vs ELTL), Basics of Data Management (DQ, Governance, MDM) and Cloud Fundamentals (IaaS, PaaS, SaaS)
- Implementation experience on at least two end-to-end DWH projects
- Experience in one ETL tool, DWH (Cloud Preferred) & Reporting tool
- Fair understanding of various Business Analytics concepts
- Descriptive, Diagnostics, Predictive and prescriptive Analytics

Program Duration

60-70 hrs. (approx.)

Persona

Professionals with 6–8 years of experience, including 4+ years in data engineering roles and at least 2 data warehousing or ETL implementations.

Learning Outcomes

At the end of this program, learners will be able to:

- Understand business problems and design scalable, cloud-ready data architectures.
- Map business and technical requirements to modern data capabilities across storage, processing, and analytics layers.
- Design efficient data pipelines using ETL, ELT, and reverse ETL; and choose the right solution between data lake, warehouse, or lakehouse.
- Implement basic security and governance practices, including authentication, authorization, and data-level access control.
- Apply data quality, observability, and governance principles to ensure trusted, reliable data ecosystems.
- Estimate capacity and cost for cloud-based data solutions using modern pricing tools.
- Confidently participate in architectural discussions by connecting technical decisions to business value.

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Program Coverage: Level 1 – Aspiring Data Architect

Introduction to Modern Data Modelling
- Principles of data modelling in the era of cloud & AI.
- Entity-relationship diagrams and semantic/knowledge graphs.

Evolving Architectural Concepts in Data
- Multi-layered architectures, data fabric, and data mesh basics.
- Importance of architecture governance in hybrid & multi-cloud setups.

Database Types and Modern Use Cases
- SQL, NoSQL, NewSQL, vector databases, and time-series DBs.
- Choosing the right database for AI/ML and real-time workloads.

Fundamentals of ETL, ELT, and Reverse ETL
- ETL vs. ELT in the cloud-native era.
- Reverse ETL and operational analytics.

Serverless & Event-Driven Data Architectures
- Benefits of serverless and event-driven designs.
- Real-world case studies in connected data systems.

Data Cleansing and Data Observability
- AI-assisted data cleansing.
- Data quality monitoring and observability tools.

Data Ingestion Essentials
- Batch, streaming, and change-data-capture (CDC) techniques.
- Handling structured, semi-structured, and unstructured data.

Modern Data Processing
- Batch vs. stream vs. micro-batch.
- Frameworks like Spark, Flink, and Kafka Streams.

Dashboarding, Data Visualization & Storytelling
- Designing intuitive dashboards with self-service BI.
- AI-driven insights and natural language querying.

Security & Privacy Fundamentals
- Zero-trust principles in data projects.
- Basics of data encryption, masking, and compliance (GDPR/CCPA).

Platform Engineering Basics for Data Architects
- Role of platform engineering in scalable data solutions.
- Leveraging containerization and Kubernetes.

Data Quality & Governance
- Establishing quality metrics with observability.
- Governance fundamentals: lineage, cataloging, and stewardship.

Introduction to Data Lakes, Data Warehouses & Lakehouses
- Evolution from warehouse → lake → lakehouse.
- When to choose which.

Introduction to Cloud Data Solutions
- Exploring AWS, Azure, GCP, Snowflake, Databricks.
- Basics of cloud-native storage & compute.

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Design 1B:

Aspiring Data Architect Level 1 for a mixed cohort of Data Scientists, Applied ML Engineers,

AI Specialists, Data Engineers Tech Leads, Data Governance to build shared data literacy.

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Context: Aspiring Data Architect

Context

The program is designed to address the evolving demands of the modern data ecosystem, guiding participants to move beyond role-specific expertise toward a systems-level understanding of data architecture. In today's enterprise landscape, data serves as a universal language that connects disciplines such as engineering, visualization, governance, and data science—each contributing a vital piece to the larger data puzzle. Success now depends on recognizing these interdependencies and collaborating across boundaries.

The program builds shared data literacy across roles, enabling participants to think beyond silos and approach architecture with a holistic business perspective. For example, data engineers learn to account for governance and visualization needs; governance specialists focus on enabling agility rather than constraining it; and data scientists align their models with practical data pipelines and business storytelling.

Structured across three layers—**Foundation**, **Deep Dive**, and **Introspect**—the program explores modern data architecture paradigms such as data mesh, data fabric, governance, lifecycle management, and case-based application. Its design cultivates systems thinking, a consulting mindset, and critical inquiry—core capabilities for data architects who must balance trade-offs and align technical decisions with business outcomes.

Program Duration

100-110 hrs. (approx.)

Persona

Data Scientists, Applied ML Engineers, AI Specialists, Data Engineers Tech Leads, Cloud Engineers, Platform Engineers, BI Developers, Data Analyst, BI solution architect, ETL Developers, Data Warehouse specialist, Data Governance with average of 4-12 years of experience

Learning Outcomes

At the end of the intervention, the learner is expected to be able to:

- Develop strong foundations in data architecture and modelling principles, including SQL/NoSQL database implementation for scalable, robust data solutions.
- Design and implement efficient data processing workflows using batch and stream processing pipelines, along with ETL/ELT workflows, to handle diverse data sources efficiently.
- Establish data governance and quality standards by creating frameworks for data integrity, governance, and observability to ensure reliable ecosystems.
- Apply advanced data modelling and automation techniques through normalization/denormalization and structured decision frameworks to evaluate trade-offs and build automated pipelines.
- Optimize storage and real-time processing through deduplication, compression, and real-time processing strategies for cost-effective solutions.
- Elevate participants from senior developer/tech lead roles to technical leadership by designing scalable, intelligent, and governed data platforms. Focus on advanced data modelling, modern architectural patterns (Lakehouse, Mesh, Fabric), and automated, policy-driven pipelines

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Program Coverage

Modern Data Architecture Foundations
- Focus on scalable, secure, and business-aligned solutions.
- Embrace cloud, AI, and distributed architectures for agile transformation

Modern Data Architecture Patterns and Design Principles
- Patterns for scalability, performance, governance, and agility.
- Storage paradigms: lake, warehouse, lakehouse, mesh, fabric, virtualization.
- AI/ML/GenAI integration in modern data architectures.

Data Mesh & Data Fabric – Modern Integration Paradigms
- Data Mesh: decentralized ownership and "data as a product."
- Data Fabric: AI/ML-driven metadata and automation for integration.

Serverless and Data Processing
- Scalable, cost-efficient serverless computing.
- Event-driven tasks via Lambda, Azure Functions, GCP Functions.
- Serverless infra considerations for AI/ML/agentic pipelines.

Data Quality, Governance, and Observability
- Data quality dimensions: accuracy, completeness, timeliness.
- Governance frameworks: cataloging, lineage, compliance (GDPR, CCPA, HIPAA).

Visualizing and Communicating Data Insights
- Empower decision-making with BI tools (Power BI, Tableau, Looker, Superset).
- Semantic Layer: enabling governed self-service and consistent metrics.

Security and Trust in Data Projects
- Security fundamentals: authentication, encryption, network isolation.
- Zero Trust and least privilege enforcement.
- Integration of cataloging and MDM tools.

Architecture Principles and Fault-Tolerant Design
- Automated failover and chaos engineering.
- Resilience patterns with Azure, AWS, GCP.

Essentials of MDM in Data Architecture
- Ensuring clean, unified data across systems.
- Role of MDM as a trusted enterprise source

Advanced Data Modelling and NoSQL vs SQL Implementation
- Multi-model architecture for structured and semi-structured data.
- Polyglot persistence: choosing relational vs. NoSQL.

Streaming Data Pipeline with Kafka
- Designing real-time streaming architectures.
- Comparing batch, micro-batch, and continuous modes.
- Event-driven design: topics, partitions, producers, consumers.

Case Study Sessions: Architecting Complex Data Solutions
- Analyze cross-industry architectures (finance, healthcare, retail).Successes and failures in data projects.
- Trade-offs: business vs. technical feasibility.

Advanced FinOps for Data Architecture
- Principles of FinOps in data-heavy environments.
- Managing cloud spend on compute, storage, transfer

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Program Coverage

Advanced Security and Disaster Recovery

- Encryption, RBAC, ABAC, PBAC models.
- Integrating cataloging and data quality.
- Continuity planning for uptime and data integrity.

Data Pipeline Orchestration and Automation (NiFi / ADF)

- Automating ETL vs. ELT workflows.
- Error handling, monitoring, and operational efficiency.

DataOps and CI/CD for Data Pipelines

- DataOps automation and CI/CD for workflows.
- DevOps practices: IaC, version control, rollback readiness.

Cloud-Native Data Solutions & Monitoring

- Cloud-native integration of batch and streaming.
- Trade-offs: cloud vs. on-prem.
- Monitoring, anomaly detection, and observability.

AI in Data Architecture

- Data pipelines and infra for AI readiness.
- Building, deploying, and monitoring AI systems.
- Using AI for pipeline generation, metadata mgmt, anomaly detection, and BI/NLP.

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Design 2:

Aspiring Data Architect Level 2 for Data Architects New to the Role

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Context: Level 2- Data Architect New to the role

Eligibility

- Proficiency in complex data modeling scenarios (e.g., many-to-many relationships, late-arriving dimensions/facts)
- Advanced knowledge of SCD types (1–6) and data ingestion patterns (batch, real-time, streaming, CDC, replication)
- Familiarity with data quality (DQ) attributes (correctness, completeness, accuracy, timeliness, integrity)
- MDM expertise (registry, consolidation, co-existence, transactional models)
- Exposure to NoSQL modeling, time-series analysis, and streaming analytics (Kafka, Kinesis)
- Knowledge of data governance tools (Collibra, Alation, Purview) and data cataloging solutions
- Experience as a Data Architect on at least two data management projects

Program Duration

60-70 hrs. (approx.)

Persona

Mid-career professionals with 8–10 years who have recently transitioned into a data architect role. They have a solid foundation in data engineering, substantial data modelling, ETL, and data governance experience.

Learning Outcomes

At the end of this program, learners will be able to:

- Define a future-ready data architecture strategy and roadmap aligned to business goals.
- Establish and operationalize a data governance model with clear ownership and accountability.
- Implement advanced security, privacy, and compliance controls across data platforms and pipelines.
- Design and optimize real-time, AI-driven, and automated data solutions using modern cloud-native tools.
- Apply FinOps and observability practices to improve cost efficiency and performance.
- Lead data architecture discussions and guide teams in designing complex, scalable enterprise data systems.

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Program Coverage: Level 2 – Data Architect New to the role

Advanced Data Modelling Techniques
- Normalization, denormalization, dimensional modelling.
- Knowledge graphs, vector embeddings, and semantic data models.

Business vs. Technical Trade-offs
- Cost-performance-security trade-offs.
- Aligning architecture with business strategy and AI adoption.

Automation & Intelligent Data Engineering
- Automating pipelines with CI/CD, orchestration, and IaC.
- AI-driven automation for pipeline optimization.

MLOps, DataOps & AIOps
- Integrating ML lifecycle with data workflows.
- AI-driven monitoring, observability, and drift detection.

Advanced FinOps for Data Architecture
- Cloud cost governance frameworks.
- Optimizing spend with autoscaling, spot instances, and storage tiering.

Case Studies in Modern Data Architecture
- Building real-world architectures with Snowflake, Databricks, BigQuery.
- Failures and learnings in enterprise adoption.

Real-time & Predictive Analytics
- Real-time streaming frameworks (Flink, Kafka).
- Predictive & generative AI-powered analytics.

Advanced Security & Compliance
- Tokenization, differential privacy, and homomorphic encryption.
- Evolving compliance frameworks for AI data pipelines.

Resilience & High Availability in Data Systems
- Multi-region replication & disaster recovery patterns.
- Designing for resilience in distributed systems.

DevOps & Data Engineering Convergence
- CI/CD for data pipelines.
- Infrastructure as Code (Terraform, Pulumi).

Cloud-Native & Emerging Data Tech
- Cloud-native databases, serverless warehouses, vector DBs.
- Data fabric, mesh, and federated query engines.

Data Storage Optimization & Sustainability
- Data deduplication, tiered storage, compression.
- Sustainable data architecture & green IT practices.

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Design 3:

Level 3 for Practitioner Data Architect (Advanced)

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Context: Level 3 for Practitioner Data Architect

Eligibility

- Complex scenarios in data modelling (M-M, Late arriving Dim/Facts)
- SCD Type 1 to 6
- Data Ingestion patterns (Batch, real-Time, Streaming, CDC, Replication)
- DQ - Correctness, Completeness, Accuracy, Timeliness & Integrity
- MDM - Registry, Consolidation, Co-Existent & Transaction style
- Other DM domains Privacy, Integration, Security, Risk Management, Policies & Procedures, ILM, Ref & Metadata
- Experience in NoSQL modelling and Time Series analysis Management
- Good knowledge to streaming analytics (Event Streams Kafka, Kinesis)
- Exposure to Data Catalogue solutions and importance
- Experience in working as a Data Architect in at least 2 DM projects
- Data Governance tool exposure - Collibra, Alation, Purview, Informatica

Program Duration

60-70 hrs. (approx.)

Persona

Professionals with 10+ years of experience and having good knowledge and understanding of the DWH and ETL design patterns and Data modelling techniques. Good experience on Data Governance

Learning Outcomes

At the end of this program, learners will be able to:

- Design Scalable and Efficient Data Architectures: Apply advanced data modeling techniques and evaluate modern architectural patterns (Data Mesh, Lakehouse, Data Fabric) based on business context and scalability needs.
- Architect Secure, Automated, and Governed Data Pipelines: Integrate DevOps, DataOps, and MLOps principles to streamline data pipelines, enforce access controls, and ensure data quality, versioning, and compliance in cloud-native environments.
- Make Informed Trade-offs in Data Solutioning: Use structured frameworks to evaluate trade-offs between business value and technical feasibility, incorporating performance, cost, and risk considerations in architectural decisions.
- Implement FinOps and Storage Optimization Strategies: Apply cost optimization techniques, understand cloud pricing models, and make data storage choices (formats, duplication, compression) that balance performance and efficiency.
- Demonstrate Real-world Architecture Thinking: Analyze case studies, collaborate on solution design, present architecture decisions, and articulate architecture narratives that reflect sound judgment and stakeholder alignment.

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Program Coverage: Level 3 for Practitioner Data Architect

Architecture Foundations & Modern Patterns
- Architecture layers, integration styles (ETL/ELT), latency vs. consistency, CAP theorem, evolution from DWH → lake → lake house → mesh/fabric

Advanced Data Lakehouse Engineering
- Delta/Iceberg/Hudi, medallion architecture, versioning, performance tuning

Data Mesh – Principles & Practices
- Domain ownership, data as a product, federated governance

Data Fabric – Architecture & Automation
- Metadata, semantic layers, virtualization, AI/ML for discovery

Strategic Architecture Paradigms
- Composability, modularity, platform thinking, change resilience
- Federated data platform for global retail

Architecture Decision Frameworks
- ADRs, ADBs, decision templates, fitness functions

Enterprise Blueprinting
- Greenfield/brownfield, rollout sequencing, reference architectures
- Data modernization for global Bank/CPG/Retail Enterprise

Cloud & Multi-Cloud Engineering
- Vendor-agnostic design, FinOps, performance, resiliency
- Multi-cloud for pharma firm

Real-Time & Streaming Architecture
- Lambda/Kappa, schema evolution, low-latency design

Performance Engineering at Scale
- Cost planning, caching, skew handling, compute-storage split

Advanced Data Governance Engineering
- Metadata, lineage, access control, decentralized policies

Data Quality & Observability for Architects
- SLAs, anomaly patterns, observability layers

ML/AI-Integrated Architecture
- Feature stores, MLOps, GenAI-assisted automation

Explainable / Responsible AI for Architects

Reading materials: Ethical Foundations & Regulations Overview of AI ethics principles (fairness, accountability, transparency) Key privacy & compliance frameworks (GDPR, CCPA, EU AI Act) Explainability Techniques
- Using cases/scenarios -- Bias Detection & Mitigation
- Fairness metrics (demographic parity, equal opportunity)
- Data-centric bias audits and synthetic data techniques
- Mitigation strategies (re-sampling, adversarial debiasing)

Architecting for Regulatory Compliance
- PII flow, auditability, zero-trust architecture

Case Study – Regulatory-Compliant Design
- Insurance/Healthcare firm with regulatory mandates (LIE-LILY)

Case Study – Cloud Migration/ Modernization Blueprint
- Rethinking data pipelines and analytics layers, not just lift and shift. Comparing Azure and AWS services, considering the move of analytics workloads from Azure to GCP. Even with seemingly similar cloud services, underlying architecture, performance models, and characteristics differ significantly.
- Modernization should include real-world cases demonstrating architectural considerations and judgment. Focus on migration from on-prem DWH to cloud-native solutions.

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Design 4:

“Architecting for Impact” for Practicing DAs

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Context: Architecting for Impact for Practising DAs

Eligibility

- Strong knowledge in database systems, data modelling (entities, attributes, relationships, normalization), and data documentation.
- Extensive experience in aligning data strategies with business outcomes
- Proficiency in SQL and familiarity with data technologies, including DWH solutions and cloud storage.
- Strong understanding of data integration, ETL processes, and data governance and standards.
- Understanding of data privacy, security, and relevant regulations (GDPR).
- Demonstrated analytical mindset and ability to communicate data concepts effectively.
- Deep understanding of modern data stack and emerging technologies (e.g., AI/ML, IoT)

Program Duration

70-80 hrs. (approx.)

Persona

Senior Data Professionals with 10–14 years of experience, transitioning into architecture roles with a focus on strategic data alignment, business benefits, and technical stack assembly. Take on more complex and strategic roles within their organizations.

Learning Outcomes

- Architect scalable, efficient data platforms, deploying best practices in data management.
- Design data models that align with organizational strategies and support scalability.
- Develop a unified Data Architecture blueprint to bridge disparate systems.
- Establish enterprise-wide standards for data consistency and quality.
- Communicate complex data concepts to stakeholders, enhancing cross-team collaboration.
- Implement data governance policies aligned with organizational goals.
- Ensure compliance with data privacy regulations through secure architecture design.

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Program Coverage: Architecting for Impact" for Practicing DAs

Foundations and Basics
- Data Models (Data Dictionary, Data Flow Diagram) CDM, LDM, and Master Data Management
- Entity Relationship Diagrams (ERDs) 101s
- Database Schema Designs and Data Platform Document

Advanced Modelling & Blueprinting
- Data Architecture Blueprint: Finalize LDM, Start PDM, Complete Data Dictionaries, start Data Flow Diagrams, Data Process Matrix and Drafting high-level overviews of the Architecture

Integration, Communication, & Standards
- Data Integration, Quality Specifications and Metadata Repository Plan
- Project Execution Excellence for Data Architects
- Data Standards Document: Begin drafting Data Sharing Agreements

Governance, Products, & Data Security
- Data Governance Strategy and Data Product Definition Document
- Data Security & Compliance Designs: Understand PII, GDPR, Design Strategies for Data Protection

Finalizing Designs & Compliance
- Database Schema Designs
- Data Security & Compliance Designs: Finalize Data Protection designs

Metadata Repo and FinOps
- ERDs (201s) and Metadata Repository Plan
- FinOps & Cloud Cost Mgmt. for Data Architects

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Design 5:
Architecting for (Data) Experiences

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Context: Architecting for (Data) Experiences

Eligibility

- Extensive experience in designing data architectures that advance business objectives & enhance efficiency.
- Deep expertise in the modern data stack (cloud data warehouses, data lakes, real-time processing).
- Proficiency in AI/ML, IoT etc. with an understanding of their impact on data architecture.
- Strong skills in enterprise data lifecycle management, ensuring data accuracy, compliance, and readiness.
- Background in data governance, policy creation, and implementation.
- Ability to communicate complex technical concepts to non-technical audiences.
- Demonstrated experience in customer journey mapping, value stream analysis, and Big Data technologies.

Program Duration

70-80 hrs. (approx.)

Persona

Senior Data Architects with 15+ years of experience in data architecture, engineering, and governance. These experts excel in strategic decision-making, complex implementations, and C-suite collaboration on data-driven initiatives. They effectively combine technical expertise with business acumen to align data strategies with organizational objectives.

Learning Outcomes

- Develop data architectures that drive tangible business outcomes and ROI, leveraging platform thinking and modern data stack.
- Align data architectures with customer journeys, value streams, and key performance metrics.
- Design and implement AI-ready data ecosystems supporting decision intelligence and enterprise-wide data lifecycle management.
- Ensure compliance with data ethics, privacy, & global regulatory standards.
- Enhance consulting and systems thinking skills to manage complex data environments.
- Strengthen skills in stakeholder influence, including persuasion, negotiation, and conflict management.
- Cultivate thought leadership and build collaborative relationships to lead successful data architecture initiatives.

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Program Design: Architecting for (Data) Experiences for Practicing DAs

AN NIIT VENTURE

Reimagine Data Architecture: Achieving Business Outcomes leveraging data

- Positioning data architecture as a strategic enabler of business transformation.
- Aligning architecture with enterprise KPIs, regulatory goals, and innovation agendas.

Platform Thinking & Data Platforms

- Architecting enterprise-scale platforms for multi-domain integration.
- Designing ecosystems for interoperability, extensibility, and vendor neutrality.

Enterprise Data Lifecycle

- Establishing lifecycle frameworks: creation, usage, governance, archival, and retirement.
- Optimizing data lifecycle for compliance (GDPR, HIPAA, CCPA) and sustainability.
- Designing lifecycle automation using metadata-driven policies.

The modern data stack

- Evaluating modern stack components: lakehouse, mesh, fabric, observability layers.
- Leveraging AI/ML capabilities in the stack (GenAI copilots, automated lineage, observability)

Getting data ready for AI

- Architecting AI-ready data pipelines with scalability, lineage, and bias monitoring.
- Managing unstructured/multi-modal data for advanced AI and GenAI use cases.
- Aligning data strategy with AI governance frameworks (responsible/ethical AI).

Basics of Data Governance

- Beyond basics: evolving governance into a value-enabling function (not compliance-only).
- Designing federated governance models (central + domain ownership).
- Embedding governance into DevOps and DataOps workflows.

Business acumen & Smart decisions

- Bridging technical architecture decisions with business strategy.
- Frameworks for quantifying ROI of architectural choices.

Customer Journey & Value Stream

- Mapping customer and value streams into architectural blueprints.
- Designing data flows aligned with customer experience and personalization.
- Using architecture to accelerate time-to-value in customer-facing initiatives.

Metrics, KPIS, GQMM

- Defining enterprise-wide metrics for data effectiveness (governance, quality, monetization).
- Linking architecture KPIs directly to board-level business metrics.

Big Data Technologies (Multi- structured)

- Strategies for handling multi-structured and streaming datasets.
- Emerging tech: vector databases, multi-modal storage, and GPU-based processing.

Data from Edge to Cloud & Automation

- Architecting unified pipelines spanning IoT, edge, and cloud.
- Balancing latency, cost, and resilience in edge-to-cloud designs.
- Automation frameworks for edge data ingestion and governance.

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Visualization & Data Storytelling
- Architect's role in enabling enterprise-wide storytelling frameworks.
- Designing semantic layers and data products for democratized insights.

Meta-data, Data Catalog, MDM, Security
- Architecting metadata-driven ecosystems for automation and discovery.
- Scaling MDM across distributed and global enterprises.
- Future-proofing security: zero trust, confidential computing, and homomorphic encryption.

Data Architecture for analytics & decision intelligence
- Architecting platforms for augmented decision-making.
- Embedding predictive, prescriptive, and agentic AI in DI ecosystems.

Consulting Acumen
- Understand the concept of Interactional Excellence
- Interaction and Collaboration skills for Architects

Systems Thinking for managing complexity
- Approaches to analysing identified challenge from different perspectives
- Developing and structuring arguments and arriving at conclusions
- Understanding interconnected nature of systems and cause effect relationships

Persuasion and Influencing Skills
- Difference between Persuasion & Influencing
- Why are Persuasion & Influencing important
- Types of Persuasion – Ethos (Credibility), Logos (Logic), Pathos (Emotion)
- Ways to overcome resistance

Negotiations and conflict management for Architects
- Success Criteria for effective Negotiation & Conflict Management
- Types of Negotiations
- 7Elements of Effective Negotiations
- Key Skills for Negotiation & Conflict Management
- How to Manage Conflict

Developing perspectives and thought leadership
- Importance of developing perspectives
- Building architecture relevant Perspectives
- Introduction to Thought Leadership
- What does it take to be a visionary, a change agent, a life-long learner, and an effective communicator?
- Continuing professional development and helping others grow

Building Relationships and Collaboration
- Importance of Teamwork
- 3 Horizons model & Stakeholder transformation
- Significance of Trust
- High Performance team framework – PERILL
- Team Vs Teaming
- Fusion Teams

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Design 6:

Level 3 for Enterprise Data Architect

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Context: Level 3 for Enterprise Data Architect

Context

The Enterprise Data Architecture Intervention program is meticulously designed around three central themes: Data Thought-Leadership, Enterprise Data Landscape, and Data Strategy, ensuring a comprehensive understanding, and empowering participants to master the intricacies of data architecture at an enterprise level.

Each themes are split into

- **Self study guides-** Self-Study Guides are designed for self-paced learning, providing comprehensive coverage of both theoretical foundations and practical applications.
- **Masterclasses-** Masterclasses are 90-minute intensive sessions led by industry experts, offering deep dives into specialized topics, and showcasing the latest trends and techniques.
- **Deep dives-** Interactive case-based workshops that simulate real-world challenges to enhance participants' problem-solving skills, critical thinking, and decision-making abilities

Participants will be asked to use the Business Data Brief to explore the scenario and present their point of view in a poster at the end of the program.

Program Duration

60-70 hrs. (approx.)

Persona

Professionals, averaging 18 years of experience, are continually driven by the need for innovation, playing a pivotal role in collaborating closely with key business and technology stakeholders, and those responsible for addressing distinct business requirements by devising and delivering tailored, high-quality solutions that meet the specific needs of the organization.

Learning Outcomes

The Aim of this program is to enable participants to:

- Handle the vast volumes of data produced by IoT devices, innovating data warehousing techniques, and using data strategically to drive business transformation.
- Revisit and refine past architectural choices and decide strategically whether to develop solutions in-house or procure them externally.
- Emphasize advancing skills in project management and interpersonal communication, such as stakeholder management and persuasive storytelling.
- Prepare these architects for leadership positions and establish them as thought leaders in their field and enable them to develop expertise across various aspects of enterprise data architecture.

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Program Coverage: Enterprise Data Architect

Data Value Chain Framework
Understand how data creates business value through its lifecycle, from creation to consumption. Learn how the data economy is shaping industries and how feedback loops enhance data-driven outcomes.

Data Ecosystem Roles
Explore the key roles in data management—from stewards and owners to governance leaders—and how they collaborate to deliver enterprise data value.

Data Observability
Learn to monitor, measure, and optimize data systems using modern observability tools and practices to ensure reliability, performance, and insights.

Multi-Structured Data
Discover techniques and tools to manage structured, semi-structured, and unstructured data using effective data modeling approaches.

Data Security Trends
Gain insights into evolving enterprise data security models, IAM requirements, and reference architectures that safeguard information.

DataOps (xOps)
Understand how DataOps and AIOps streamline operations, ensure data quality, and drive continuous improvement across the data lifecycle.

Information Consumers
Identify who uses enterprise data, how they access it, and how information architecture supports diverse analytical needs.

Information Personalization
Learn how to tailor data experiences for different personas, enabling contextual insights and more relevant decision-making.

Emerging Data Trends
Stay updated on new technologies, methodologies, and regulations driving the evolution of the modern data landscape.

Data Integration Trends
Explore integration tools and models that connect data from diverse sources, ensuring unified and reliable analytical solutions.

Data Governance
Understand the pillars of data governance, how to implement them effectively, and how they ensure trusted, well-managed enterprise data.

Data-led Decision-Making
Learn how data, analytics, and AI/ML empower decision-makers to make faster, evidence-based business choices.

Effective Visualizations
Discover how to create meaningful visualizations that make complex data easy to interpret and act upon.

Data Storytelling
Master the art of turning data insights into compelling stories that inspire understanding and action.

Optimal Data Security
Learn to design secure data environments, assess risks, and implement proactive strategies to strengthen data protection.

Business 'Uses' of Data
Understand how different business personas use data and how to align data access and personalization with user needs.

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Program Coverage: Enterprise Data Architect

Business 'Value' of Data
Learn how to measure, manage, and maximize the value of enterprise data through its lifecycle from discovery to delivery.

Biz-aligned Data Strategy
Explore how data architecture connects with business, application, and information strategies to build agile enterprises.

Business-IT Strategy
Understand how business and IT strategies align through data-driven planning to power enterprise transformation.

Data Maturity Model
Learn how to assess and improve data maturity using models that benchmark progress and guide strategic development.

Introduction to Data Strategy
Build a foundational understanding of data strategy—its elements, roadmap, and alignment with business goals and architecture.

Business Acumen for Data Professionals
Strengthen your understanding of how business data powers operations, strategy, and decision-making across the enterprise.

Data Compliance | Standards
Gain a clear understanding of why data standards are essential for consistency, interoperability, and trust. Learn about key global and emerging standards shaping data governance, and explore the ethical and compliance aspects of AI and ML adoption.

Data Architectural Decisions
Learn how architects make data-driven decisions, balancing usability, performance, and strategy while communicating through storytelling

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StackRoute's Approach and Guidelines to Program Delivery

We follow two methods to deliver our interventions:

- Short term deep skilling programs | daily sessions | technical hands-on | delivered for 10-15 days
- Long term transformative journeys | sessions 2-3 days a week | less than 10 hrs commitment a week

Program pre-requisites and guidelines | Program Flyers and Info-session for business stakeholder and learner buy-in | Validation of Learner Persona | Tight co-ordination with L&D and Business for monitoring, tracking and alerts

It is StackRoute's responsibility for end-to-end design and delivery of the intervention. This includes selection of mentors, content curation/creation and alignment with learning outcomes, design of rubrics for introspects (assessments), and post program learning impact analysis.

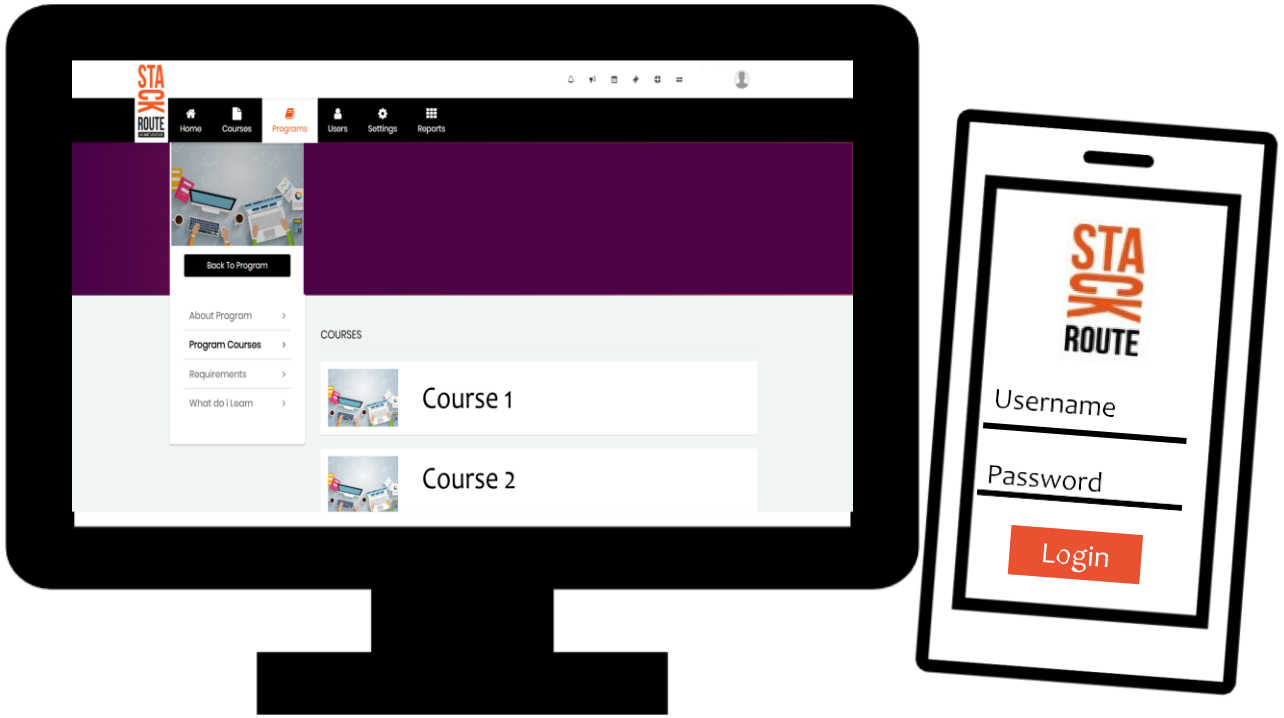
Our mentors are senior practitioners who have their consulting practice. Mentors chosen for a particular program will be available for introductions and detailed discussions (not evaluations) with business stakeholders.

An overall approach to sessions will be presented by the mentors in discussion with the business stakeholders. The design of transformative programs is to spark discussion, encourage questioning, and promote active engagement and hence the presentations or content need not be self explanatory. Due to this reason, mentors will present the approach in a meeting with business stakeholders and not share the content for any review or evaluation unless a specific module or a certain coverage demands so.

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Digital Infrastructure

Learners can access learning content to stay on the right track of their learning journey



- Integrated with Gototraining and Zoom
- Access learning platform from any mobile device
- Access course content and pre-reads on the go
- Track progress any time, anywhere
- A calendar which highlights the upcoming action items and sessions
- Share session feedback and your inputs for sessions
- Take assessments and evaluations on the go

Access LMS any time and any place

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Next Steps

Note- the design has been customized based on the inputs received. We will fine tune the design over subsequent conversations.

Discussion with the Business Team

- Showcase the design to business team and relevant leaders.
- Understand specific needs from business team including learner job/titles, work ex, nature of work and the intended outcomes of the intervention.
- Understand business needs both short term and long term.
- Understand learner commitment (time available).

Post which:

- SR will propose first level design and finalize the design with discussions
- Finalize on the execution plan.

While all learning interventions are customized, any specific requirements that can come from a cohort that is heterogenous in terms of work experience, roles, and designations may require ongoing customization throughout the duration. In such cases, the program will no longer fall under the purview of a training program and will be scoped further by SR Consulting.

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Thank You!

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